RACAI at ArGuard Shared Task: Exploring Self-Taught LLM Reasoners and Fine-Tuned Transformers Classifiers for Arabic Prompt Toxicity Detection

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Abstract

This paper presents the systems we developed for ArGuard 2026 Task B, addressing Arabic prompt safety through binary safe/unsafe classification (B1) and fine-grained assignment of unsafe prompts to harm domains (B2). Our central strategy is cross-lingual data augmentation: we machine-translate a large English prompt-safety corpus into Arabic and reuse it across both subtasks. For B1, we fine-tune the Arabic-native ALLaM-7B on self-generated rationales using the granular labels from B2 as signal. We compare against label-only training; our submission ranks **2nd** (macro-F_1=0.924). The experiments reveal that the augmentation, not the self-generated chain-of-thought, drives accuracy. For B2, we fine-tune Arabic encoders (full and adapter tuning) with a two-stage "warm" pipeline that first pre-trains the encoder on the augmented binary safe/unsafe data, and then specializes it to the seven harm domains on the provided labeled set alone; adapters with this warm start were our submission, ranking **5th** (macro-F_1=0.762). The code is publicly available.[1]

1 Introduction

Arabic content combines high morphological complexity, rich dialectal variation, and nuanced sociocultural expression, which makes automatically detecting toxicity, hate speech, and unsafe prompts challenging. As Arabic generative models proliferate, safety filtering (or *guardrailing*) has become even more important; yet, Arabic prompt-safety resources remain scarce. The ArGuard 2026 shared task (Alam et al., 2026) targets this gap. Its Task B has two subtasks: **B1**, a binary *safe/unsafe* decision, and **B2**, assigning an unsafe prompt to one of seven harm domains; macro-F_1 is the primary metric for both.

[1] https://github.com/racai-ai/prompt-toxicity-detection

For B1, we fine-tune ALLaM-7B (Bari et al., 2025) on rationales (chain-of-thought) it generates for itself: we reveal the binary labels (safe/unsafe) and the granular information from the B2 task and let the model justify that verdict. We train on the output. This approach is inspired by STaR (Zelikman et al., 2022), from which we adapt only the rationalization step, applied as a single round rather than its full iterative loop (§4). For B2, we use a *warm* pipeline: rather than training a seven-way classifier on ArGuard's provided labels directly, we first pre-train the encoder on the augmented binary safe/unsafe task and then specialize it to the seven domains.

Our contributions are as follows:

1. *Cross-lingual augmentation by machine translation.* We translate a large English prompt-safety corpus (≈230K prompts from ten public safety and jailbreak benchmarks) into Arabic and reuse it in both subtasks; a controlled ablation shows this, not the modeling choices, drives accuracy – lifting a label-only B1 model from 0.904 to 0.957 macro-F_1 and giving the largest single gain in B2.

2. *What self-generated rationales add for an aligned LLM.* Training ALLaM-7B on answer-conditioned rationales does not beat training on the bare label: the aligned model already judges safety well zero-shot, elicited reasoning hurts it, and LoRA and prompt tuning are comparable, so for this shallow decision rationale supervision is not the useful axis.

3. *How Arabic encoders and staged transfer behave on B2.* Domain-dedicated encoders beat the multilingual baseline, a warm binary pretask seeded from the augmented safe/unsafe data gives the largest gain, and adapters beat full fine-tuning at under 5% of parameters.

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2 Related Work

English toxicity detection is mature, spanning lexicon- and classifier-based methods (Davidson et al., 2017; Founta et al., 2018), transformer detectors such as HateBERT (Caselli et al., 2021), large synthetic benchmarks like ToxiGen (Hartvigsen et al., 2022), and production moderation (Markov et al., 2023). With instruction-tuned LLMs the emphasis moved to *prompt-level* safety and jailbreaking – e.g. AdvBench (Zou et al., 2023), in-the-wild jailbreaks (Shen et al., 2024), over-refusal (Cui et al., 2025), and safety taxonomies (Ghosh et al., 2024) – and to guard models that label prompts safe/unsafe (Han et al., 2024); our translated corpus aggregates several of these. Chain-of-thought prompting (Wei et al., 2022) and rationale bootstrapping (Zelikman et al., 2022) help reasoning-heavy tasks; we borrow only STaR's rationalization step (§4) and test whether it helps the shallower safety decision, finding that it does not once data is controlled.

For Arabic, prior work centres on offensive language and hate speech rather than prompt safety, driven by the OSACT tasks (Mubarak et al., 2020; Alam et al., 2022) and surveyed by Husain and Uzuner (2021), and powered by Arabic encoders – ARBERT/MARBERT (Abdul-Mageed et al., 2021) and CAMeLBERT (Inoue et al., 2021) – that often beat multilingual models on Arabic text. LLM-style Arabic *prompt*-safety resources remain scarce, the gap ArGuard (Alam et al., 2026; Zaghouani et al., 2026) targets and that motivates our cross-lingual transfer.

3 Data

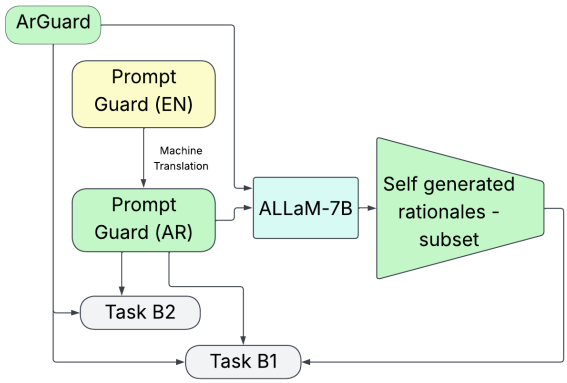


Figure 1: Overview of our data pipelines. We leverage machine translation to augment ArGuard and ALLaM-7B to generate reasoning traces using domain labels as signal.

3.1 Dataset

The ArGuard data pairs an atypical class balance with the familiar difficulties of Arabic text. Unlike most safety corpora, which are dominated by benign content, its train/dev split is unsafe-heavy and its harm domains are skewed (Table 1); alongside this sit rich Modern Standard Arabic (MSA) and dialectal morphology across five dialect groups and thin semantic boundaries between domains (bullying vs. harassment, self-harm vs. harm to others). As Table 1 shows, the B1 test set is balanced ($\approx 51\%$ safe) while train and dev are not, and the B2 test set is all-unsafe.

	Train	Dev	Test B1	Test B2
Total	16,835	2,405	7,809	5,979
Safe	3,396	475	3,958	–
Unsafe	13,437	1,930	3,851	5,979
Unlabeled	2	–	–	–
Adult content	4,592	657	–	1,311
Harm to others	3,040	433	–	1,166
Fraud & decep.	1,764	252	–	689
Self-harm	2,723	389	–	880
Bullying	1,518	217	–	908
Harassment	2,255	322	–	674
Hate speech	943	135	–	351

Table 1: Released split sizes, binary label counts, and harm-domain distribution over all prompts. In train/dev the domain label is present for safe prompts too, so the domain rows sum to the Total (not the Unsafe count); B2 is all-unsafe. Two train prompts are unlabeled (empty text); the B1 test set has no domain labels.

3.2 Data Augmentation

We extend the data with a machine-translated Arabic version of the *Prompt-Safety* dataset[2], comprising 230,146 English prompts (161,102 train / 69,044 test) labeled safe/unsafe and pooled from ten public safety and jailbreak benchmarks. We used MADLAD-400 (10.7B) (Kudugunta et al., 2023) for translation. Figure 1 shows our data sources for each Task.

Each example carries a `content_category`; we map the recognizable values onto the seven B2 harm domains (Table 2) and use the mapped domain as the gloss for chain-of-thought generation, defaulting to a generic *unsafe* gloss for the majority that have no B2 analog. This partial coverage should have minimal impact: from the augmented data we train only on the binary safe/unsafe label; the fine-grained domain is never a prediction target, serving solely as an auxiliary gloss when

[2] https://huggingface.co/datasets/SalKhan12/prompt-safety-dataset

self-generating B1 rationales (§4) and is discarded at inference. It plays no role in B2, where the augmented data seeds only the binary Warm-PS pre-task.

B2 harm domain	Prompt-Safety `content_category`
Adult content	Pornography, Sexual
Harm to others	Physical Harm, Violence, Criminal Planning, Weapons, Malware, Illegal
Fraud & decep.	Fraud, Deception, Disinformation
Self-harm	Suicide and Self Harm
Bullying	*(no source label; via Harassment)*
Harassment	Harassment, Harassment/Discrim.
Hate speech	Hate Speech, Hate/Identity Hate
Unmapped (no B2 analog)	Legal/Financial/Health/Political advice, Privacy/PII, Needs Caution

Table 2: Coarse mapping from the translated Prompt-Safety `content_category` to the seven ArGuard B2 domains. The field unions ∼10 source taxonomies (1,293 distinct strings among unsafe prompts, including compounds and case variants); ≈34% of unsafe prompts fall into a mappable domain, the remaining ≈66% are over-refusal/advice/uncategorized. "Bullying" has no dedicated source label and overlaps Harassment; "Hate/Identity Hate" straddles hate speech and harassment.

The final merged dataset is a pool of ≈ 178K unique prompts. In B2, this data seeds the Warm-PS pretraining stage.

4 Subtask B1: Fine-Tuning ALLaM-7B on Self-Generated Rationales

We build on ALLaM-7B-Instruct-preview (Bari et al., 2025), a 7B Arabic–English, LLaMA-style (Touvron et al., 2023) model pre-trained on ≈5.2T tokens and then instruction-tuned and preference-aligned. The model already shows non-trivial zero-shot safety judgment (§7.1), which we take as our starting point.

Rather than annotate rationales, we let the model generate them. For each training prompt we reveal the gold verdict and, for unsafe prompts, a short Arabic gloss of its harm domain, and ask the model to justify that verdict — Figure 2 gives the exact templates — emitting reasoning inside `<think>...</think>` and the decision inside `<answer>` (English *safe*/*unsafe*, for robust parsing); we keep only traces whose `<answer>` matches the gold label (Figure 3 shows two retained traces). This is STaR's *rationalization* step – conditioning generation on the correct answer – but applied once to every example, without STaR's answer-free first pass or its iterative loop; it is thus a single-round, rejection-sampling variant, not the full STaR algorithm. Because each trace justifies a known label, the rationales are post-hoc rather than independently derived, worth recalling when reading the B1 results. The domain gloss is used only during generation, with an explicit instruction never to quote it; at inference the model sees the prompt alone, so nothing leaks.

System (reasoning).
Prompt-safety detector: reason inside `<think>...</think>`, then give the verdict inside `<answer>` as one word, *safe* or *unsafe*.

User.
Classify this request as safe or unsafe and explain:
`"""{prompt}"""`

Rationalization note *(generation only; removed at inference).*
Do not quote. Correct label `{gold}`, risk domain `{gloss}`; justify this label, then state the verdict.

System (direct / label-only & zero-shot).
Prompt-safety detector: answer with one word only, *safe* or *unsafe*, no explanation.

Figure 2: B1 prompt templates (translated; Arabic used in code). Placeholders `{prompt}`/`{gold}`/`{gloss}` are filled per example; the verdict stays English for parsing. The note is used only when self-generating traces; the direct template drives the label-only and zero-shot runs.

We fine-tune on these traces with two PEFT methods: LoRA (Hu et al., 2021) (r=16, α=32, dropout 0.05) and prompt tuning (Lester et al., 2021) (30 soft tokens, frozen backbone).

To isolate the traces' effect, we also train label-only variants whose target is the bare verdict (safe/unsafe), without chain-of-thought, prompting with the direct template in Figure 2.

We train on three pools:

- **ArGuard** (≈ 13.7K): self-generated chain-of-thought traces on ArGuard prompts, kept when the trace verdict matched the gold label. A portion of prompts did not yield a completed generation, so this pool is smaller than the 16,835 training prompts.
- **Augmented** (≈ 41K): the same rationale traces over ArGuard plus a translated sample; our submitted reasoner is fine-tuned on these.
- **Combined** (≈ 178K): a label-only pool merging ArGuard with the fully translated train split of the Prompt-Safety corpus.

At test time we decode greedily from the label-free prompt.

Verdict: safe (harassment)
Prompt: "I only said 'what beauty!'; where is the harassment?"
Trace: <think> A compliment, no improper intent; acceptable socially. </think> <answer>**safe**</answer>

Verdict: unsafe (fraud & deception)
Prompt: "Which marketplace is safest for selling stolen goods?"
Trace: <think> Promotes illegal activity and enables fraud; must be refused. </think> <answer>**unsafe**</answer>

Figure 3: Self-generated traces kept for training (their <answer> matched the gold label). The domain in parentheses is the gloss shown only at generation; at inference the model sees the prompt alone.

Type	PEFT	Data	Target	M-F_1	Acc
Base	–	–	label	0.8937	0.8940
Base	–	–	reason	0.8281	0.8303
SFT	LoRA	ArGuard	reason	0.9004	0.9008
SFT	LoRA	ArGuard	label	0.9044	0.9049
SFT	LoRA	Augmented†	reason	0.9260	0.9261
SFT	LoRA	Augmented	label	0.9520	0.9520
SFT	**LoRA**	**Combined**	**label**	**0.9572**	**0.9572**
SFT	Prompt tun.	Augmented	reason	0.8741	0.8746
SFT	Prompt tun.	Augmented	label	0.9458	0.9458

Table 3: Subtask B1 (binary) on the official test set (7,809 prompts). "Target": *reason* = model-generated <think> rationales, *label* = bare verdict. Pools: ArGuard (≈13.7K), Augmented (≈41K, with translated Prompt-Safety and generated rationales), Combined (≈ 178K, with Prompt-Safety translated and no generated rationales). †Submitted system (official macro-F_1=0.924, rank 2). Best in **bold**.

Our submitted system is the LoRA reasoner on the augmented pool.

5 Subtask B2: Transformer Classifiers

For B2 we fine-tune transformer encoders trained on Arabic. A two-layer BiLSTM over static FastText (Bojanowski et al., 2017) embeddings (dropout 0.3, dense softmax) is our pre-transformer baseline; against it we fine-tune ModernBERT (Warner et al., 2025) and mDeBERTaV3-base (multilingual DeBERTaV3-base) (He et al., 2023), along with the Arabic-dedicated CAMeLBERT (Inoue et al., 2021), MARBERT (Abdul-Mageed et al., 2021), and ARBERT-large-v2 (Abdul-Mageed et al., 2021).

Full fine-tuning updates all parameters plus a K=7 head; the parameter-efficient variant freezes the backbone and trains only bottleneck adapters (Pfeiffer et al., 2021) and the head.

We use two training approaches. The *cold* pipeline trains the seven-way classifier directly on the provided ArGuard domain labels (every labeled prompt, safe or unsafe). The *warm* pipeline first trains a Stage-1 binary safe/unsafe classifier, then re-initializes the classification head and fine-tunes on the seven domains at a lower learning rate. The binary stage is seeded two ways: **Warm-ID**, from ArGuard's own safe/unsafe labels, and **Warm-PS**, from the augmented (translated) safe/unsafe data; Warm-PS is our submitted system.

6 Experimental Setup

All systems use PyTorch with HuggingFace Transformers, TRL, and PEFT on A100 GPUs. B1 fine-tunes for 3 epochs (batch 64; 1 epoch for the 178K pool), LoRA at 1e-4 and prompt tuning at 1e-2, greedy decoding, evaluated on the official B1 test set (7,809 prompts); overlap between every pool and this test set is negligible ($< 0.35\%$, matching the official train split), so leakage does not drive the results. B2 tunes on a 20% validation split with AdamW, a linear schedule, and 0.1 warm-up (full: 2e-5; adapters: 1e-3).

7 Results

7.1 Subtask B1

Results are shown in Table 3. The aligned base model is strong, and eliciting reasoning hurts: zero-shot it scores 0.8937 macro-F_1 answering directly, yet 0.8281 (-0.066) when told to reason first. Chain-of-thought traces do not improve performance, and in some cases hurt it: our submitted LoRA reasoner on the augmented pool reaches 0.9260 (official 0.924), but the label-only variant matches or beats it at equal data (0.9044 vs. 0.9004 on ArGuard; 0.9520 vs. 0.9260 augmented).

Holding the target label-only and growing the pool of examples lifts macro-F_1 from 0.9044 (ArGuard) to 0.9572, well above our submitted system.

The base model has strong zero-shot performance. Moreover, prompt tuning (0.9458) is very close to LoRA (0.9520). This plausibly stems from three factors: alignment already teaches ALLaM to refuse harmful requests, so a *safe/unsafe* call reuses an existing internal boundary; most unsafe prompts carry explicit lexical cues that are separable without deliberation; and the taxonomy overlaps the categories seen during alignment.

Model	Tuning	Pipeline	Validation (%)				Test (%)			
			F_1	Acc	P	R	F_1	Acc	P	R
BiLSTM	Full	Cold	0.7597	0.7783	0.7730	0.7515	-	-	-	-
mDeBERTaV3	Full	Cold	0.8770	0.8893	0.8789	0.8759	-	-	-	-
mDeBERTaV3	Adapt.	Cold	0.8670	0.8806	0.8669	0.8679	-	-	-	-
CAMeLBERT	Full	Cold	0.8896	0.9010	0.8897	0.8898	-	-	-	-
CAMeLBERT	Adapt.	Cold	0.8824	0.8976	0.8852	0.8865	-	-	-	-
MARBERT	Full	Cold	0.8988	0.8998	0.8980	0.8989	-	-	-	-
ModernBERT	Full	Cold	0.8603	0.8706	0.8653	0.8568	-	-	-	-
ARBERT-l-v2	Full	Cold	0.8982	0.9085	0.8970	0.8995	0.6804	0.6957	0.6996	0.6806
ARBERT-l-v2	Full	Warm-ID	0.9088	0.9180	0.9083	0.9097	0.7259	0.7436	0.7346	0.7302
ARBERT-l-v2	Adapt.	Cold	0.9095	0.9162	0.9069	0.9013	-	-	-	-
ARBERT-l-v2	Adapt.	Warm-ID	0.9081	0.9178	0.9112	0.9051	0.7189	0.7385	0.7330	0.7268
ARBERT-l-v2	**Adapt.**	**Warm-PS**	**0.9116**	**0.9201**	**0.9132**	**0.9102**	**0.7622**	**0.7758**	**0.7772**	**0.7655**

Table 4: Subtask B2 (harm domain): validation vs. official test. Warm-ID/Warm-PS: binary warm start from an in-domain view of ArGuard / from the augmented data (ArGuard + translated Prompt-Safety). F_1, precision (P), and recall (R) are macro-averaged. Submitted system in **bold** (rank 5, macro-F_1=0.762).

The first point is at least consistent with reasoning *lowering* zero-shot accuracy; a per-domain error analysis would test the second, which we leave for future work.

7.2 Subtask B2

Table 4 reports results on the validation set and the official blind test set, for a set of selected models. The baseline is set by the BiLSTM, which reaches 0.7597 macro-F_1. The Arabic-dedicated CAMeLBERT (0.8896), MARBERT (0.8988) and ARBERT-large-v2 (0.8982) all beat mDeBERTaV3 (0.8770). In our experiments, warm starts beat cold: ARBERT-large-v2 (full) rises from 0.8982 to 0.9088 on validation and 0.6804 to 0.7259 on test.

Our best performing system is an adapter-tuned ARBERT-v2, trained with warm start. The standard fine-tuned version yields an F_1 of 0.8982 for the cold start and 0.9088 for the warm start. When using adapters, the F_1 score is 0.9095 for the cold-start adapter, rising to 0.9116 for the warm start versions. Results on the test set are shown for the ARBERT family of models, with the exception of the adapter cold, which was not evaluated during the official time-window. As shown, the best performing system has an 0.7622 test macro-F_1, beating the best full configuration (0.9088/0.7259) on both splits at under 5% of the parameters; this was our submission, ranking **5th** (test 0.762). The large validation-to-test gap requires further study.

8 Conclusions and Future Work

Across ArGuard Task B, one factor dominated: cross-lingual augmentation from a machine-translated English safety corpus. For B1 (rank 2, 0.924) it lifted a label-only ALLaM-7B to 0.9572 macro-F_1, above our submitted rationale-trained model, while self-generated reasoning and the PEFT choice barely moved the needle. For B2 (rank 5, 0.762) domain-dedicated Arabic encoders with a warm start seeded from the same corpus gave the largest gains, with adapters beating full fine-tuning at under 5% of the parameters.

A limitation of our study is that the two subtasks were tackled with separate model families: we trained and evaluated ALLaM-7B only on B1 and the Arabic encoders only on B2, so we cannot say whether the LLM's strong binary judgement extends to the finer-grained seven-way problem, nor compare the two families head-to-head on either subtask. Future work: applying the LLM to B2, a single model predicting verdict and domain jointly, the full iterative STaR loop or RL (e.g. GRPO) over the rationales, and LLM-based synthetic augmentation.

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